

8 (a) Striated muscle is made of many fibres. Each fibre is composed of myofibrils.

The striated appearance of the muscle fibre is due to the arrangement of two types of protein filaments, thick filaments and thin filaments, within the sarcomeres of the myofibril.

Describe the main structural features of thick filaments and thin filaments in the sarcomere.

thick filaments
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thin filaments
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(b) Describe the role of calcium ions in the contraction of striated muscle.

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(c) Muscle cells sometimes have to carry out respiration in anaerobic conditions.

Describe how respiration occurs in anaerobic conditions in a muscle cell **and** state why it is important that this process occurs.

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[Total: 13]

Section B

Answer **one** question.